

AI Robot Receptionist — Hardware Component List

Hardware-focused bill of materials for the robotics/mechatronics part of the project. RFID/NFC reader and thermal printer are intentionally excluded.

A. Main Computing & Human Interface

No.	Component	Recommended specification	Purpose
1	Raspberry Pi 5	4 GB RAM	Main computer / AI hardware
2	MicroSD Card	64 GB or 128 GB, Class 10	Raspberry Pi operating system and software
3	Raspberry Pi 5 Active Cooler	Official or compatible	Cooling
4	ESP32 Development Board	ESP32-WROOM-32	Real-time hardware control
5	7-inch Capacitive Touchscreen LCD	HDMI, 800×480 or 1024×600	User interface/display
6	USB Camera / Raspberry Pi Camera	1080p recommended	Vision and QR scanning
7	USB Microphone / Microphone Array	Noise-cancelling preferred	Voice input
8	USB Speaker / Powered Speaker	3–5 W or suitable USB speaker	Voice output

B. Robot Motion & Actuation

No.	Component	Recommended specification	Purpose
9	TB6612FNG Dual Motor Driver	2-channel	DC motor control
10	DC Geared Motors	12 V, torque/RPM selected for chassis	Robot movement
11	Robot Wheels	2 ×, matched to motors	Robot movement
12	Caster Wheel	1 ×	Mechanical support
13	Servo Motor	MG90S or MG996R as required	Head/arm/pan-tilt mechanism

C. Sensors & Safety

No.	Component	Recommended specification	Purpose
14	HC-SR04 Ultrasonic Sensor	5 V module	Basic obstacle detection
15	VL53L0X / VL53L1X ToF Sensor	I ² C	Accurate short-range distance
16	MPU6050 IMU	6-axis, I ² C	Orientation/motion sensing
17	PIR Motion Sensor	HC-SR501 or equivalent	Detect approaching person
18	Emergency Stop Push Button	Latching type preferred	Safety/manual stop
19	RGB LED / LED Strip	5 V or appropriate type	Status indication
20	Buzzer	3.3 V/5 V module	Alerts
21	Push Buttons	Momentary	Manual controls

D. Power System

No.	Component	Recommended specification	Purpose
22	Li-ion Battery Pack	Voltage selected for motors	Main portable power
23	Battery Management System (BMS)	Matched to battery configuration	Battery protection
24	DC-DC Buck Converter	Motor-battery voltage → stable 5 V	Power for Pi/electronics
25	Main Power Switch	Suitable current rating	Power ON/OFF
26	Fuse + Fuse Holder	Rating selected from load	Electrical protection
27	XT30/XT60 Power Connectors	Match battery/current	High-current connections
28	JST Connectors	Appropriate series	Low-voltage connections

E. Wiring, Electronics & Tools

No.	Component	Recommended specification	Purpose
29	Jumper Wires	M-M, M-F, F-F	Prototyping
30	GPIO / Ribbon Cable	40-pin compatible	Pi GPIO connections
31	Breadboard / Prototype PCB	Standard size	Prototyping
32	Screw Terminal Blocks	Suitable pitch/current	Secure wiring
33	Resistor Kit	Common values	LED/signal circuits
34	Heat-Shrink Tubing	Assorted sizes	Insulation/cable protection
35	Soldering Iron Kit	Temperature-controlled preferred	Wiring/assembly
36	Wire Stripper / Cutter	Basic electronics tool	Wire preparation
37	Crimping Tool	For selected connectors	Connector assembly
38	Digital Multimeter	DC voltage/current/continuity	Testing and troubleshooting

F. Mechanical Structure

No.	Component	Recommended specification	Purpose
39	3D-Printed Robot Chassis	Custom design	Mechanical structure
40	Motor Mounting Brackets	Matched to motors	Motor mounting
41	M3/M4 Screws, Nuts & Spacers	Assorted kit	Mechanical assembly
42	Cable Management	Cable ties / spiral wrap	Organized wiring
43	Rubber Feet / Vibration Pads	As required	Stability and vibration reduction

Recommended first-purchase set

Item	Reason
Raspberry Pi 5 4GB	Main computer
ESP32 Development Board	Hardware controller
7-inch HDMI Capacitive Touchscreen	Display/touch interface

Item	Reason
Camera	Vision/QR scanning
USB microphone + speaker	Voice interaction
TB6612FNG + 2 DC geared motors + wheels + caster	Movement
Ultrasonic/ToF sensors + MPU6050	Obstacle and motion sensing
Battery + BMS + buck converter + fuse + switch	Power system
Multimeter + soldering/crimping tools	Hardware assembly/testing
3D-printed chassis + mounting hardware	Robot body

Hardware architecture: Raspberry Pi 5 handles the main computer/interface functions; ESP32 handles motors and real-time sensors. Pi 5 and ESP32 can communicate over USB serial. Always select battery, BMS, fuse, buck converter, motor driver, and wiring ratings from the actual motor current and measured system load before final assembly.